

Jonathan Fabien

772-634-6519 | jonathanfabien675@gmail.com | [linkedin.com/in/jonathan-fabien](https://www.linkedin.com/in/jonathan-fabien) | github.com/PapiFab1

EDUCATION

University of Central Florida

Bachelor of Science in Computer Science

Orlando, FL

May 2026

Relevant Coursework: Data Structures & Algorithms, Object Oriented Programming, Computer Logic & Organization, Algorithms in Machine Learning, Database Systems, Enterprise Computing

TECHNICAL SKILLS

Languages: Java, Python, C, C#, JavaScript, TypeScript, HTML/CSS, SQL, PHP

Frameworks and Libraries: React, Node.js, Express, Flask, Pandas, NumPy, Bootstrap 5

Databases/ORMs: PostgreSQL, MySQL, MongoDB, Redis, Drizzle ORM

APIs and Developer Tools: Gemini API, VS Code Extension API, REST APIs, Git, GitHub, VS Code, Postman, Jira

Cloud/DevOps: AWS, Docker

EXPERIENCE

Software Engineer | *Full-Stack, TypeScript, React, Node.js, MongoDB*
COMPETE Co.

August 2025 – Present

Orlando, FL

- Hired part-time after completing a company-sponsored university capstone project with COMPETE Co.; continued developing full-stack features, debugging application issues, and implementing user experience improvements.
- Developed full-stack **team event and tournament systems** using **React**, **Node.js**, and **MongoDB**, supporting live brackets, team matchups, event status tracking, and role-based event participation.
- Built interactive React interfaces for event details, matchup previews, countdown timers, bracket progression, winner highlighting, and team lineup displays, improving event navigation efficiency by **30%**.
- Optimized API routes, controllers, and MongoDB population/update logic to keep event, team, matchup, and bracket data synchronized across frontend and backend views.
- Implemented role-aware participation logic for captains and team members, enforcing event eligibility rules based on team sport, lineup size, event status, and user permissions.

Undergraduate Research Assistant | *Python, Pandas, Data Processing, GSEA*
University of Central Florida, Supervised by Dr. Haiyan Hu

January 2024 – April 2024

Orlando, FL

- Developed a **Python** program to preprocess, clean, and analyze GSEA output files, reducing manual data-handling time by over **61%**.
- Used **Pandas** to parse structured research data, organize gene set results, filter statistical outputs, and generate reproducible analysis tables.
- Automated repetitive data transformation tasks including file parsing, threshold-based filtering, result formatting, and pathway comparison across gene expression datasets.
- Presented processed analysis results to the research team, explaining how the Python workflow improved consistency, reproducibility, and efficiency in reviewing enriched gene sets.

PROJECTS

SyntaxSense | *TypeScript, Node.js, PostgreSQL, Drizzle ORM, Gemini API, VS Code API*

- Built an AI-powered **VS Code extension** that helps developers learn unfamiliar syntax by generating contextual explanations for highlighted code or the active line across **5+** dynamically detected languages.
- Integrated the **Gemini API** with a custom extension panel to return contextual code explanations in **2–5 seconds**, reducing syntax lookup time by an estimated **80%+** compared to manually searching external resources.
- Designed a **PostgreSQL** schema with **Drizzle ORM** to store code snippets, detected language, AI explanations, timestamps, and learning history across sessions.
- Implemented an interactive quiz mode that generates multiple-choice questions from recent syntax explanations, validates selected answers, and reinforces concepts directly inside the developer workflow.

Task-It | *TypeScript, JavaScript, Node.js, Express, REST APIs, Git*

- Developed **RESTful API routes** for task management workflows, supporting task creation, retrieval, updating, deletion, and search/filter functionality.
- Built backend authentication and account recovery routes for user registration, login, forgot password, and password reset workflows.
- Wrote automated test scripts covering core **Express** API routes, increasing test coverage to over **82%** and improving backend reliability.

CERTIFICATIONS

AWS Cloud Security Foundations Certificate | *Amazon Web Services Training and Certification*

2026